

BIO1

General Biology 1

Grade 12, Semester 1 study outline

A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.

01. Cell Theory

I. Introduction

Ancient Greek philosophers such as Aristotle (384 -322 B.C.) and Paracelsus concluded that "all animals

II. Timeline of Cell Theory

- The first useful microscope called the flea glasses was invented.

III. The Cell Theory

There are three (3) postulates of cell theory.

1. All living things are made up of cells

IV. Exceptions to the Cell Theory

Cell Theory does not have a universal application. Some living organisms do not have true cells.

1. Set of Genes

2. Plasma membrane

3. Metabolic Machinery

NOTE: Viruses do not easily fit these parameters of a true cell. Because they lack plasma membrane and metabolic

1. They have genetically determined macromolecular organization

2. They have genetic hereditary material in the form of DNA or RNA

3. A capacity of auto-reproduction

4. A capacity for mutation

V. Unlocking the terms

B. DNA (deoxyribonucleic acid)

- It is a double-stranded organic chemical that contains genetic information that is passed onto offspring.

02. Cell Structure and Functions

I. Introduction

According to the cell theory plants and animals are made up of the unit called cells. As a living organism,

II. What are cells?

The actual word, cell, in turn, comes from Latin and means "a chamber." This terminology arises from the

III. Eukaryotic cells

Eukaryotic cells can be a plant or an animal cell. They differ in size, shape and some parts in the plant cell, you

IV. Parts of a cell and its function

B. RIBOSOMES- protein factories

- They are composed of ribonucleic acid (RNA) and proteins and they are the sites of protein synthesis.

C. ENDOPLASMIC RETICULUM- highway

- is literally a "tiny network present within the cytoplasm."

D. GOLGI APPARATUS- receiving, sorting, shipping center of the cell

- We can think of the Golgi as a warehouse for receiving, sorting, shipping, and even some manufacturing.

E. LYSOSOMES- digestive compartment

- contains digestive enzymes which, when released, digest or break down foodstuffs and other materials

F. VACUOLE- compartment

- it is a storage sac for various digestive enzymes.

G. CYTOPLASM- refers only to the region between the nucleus and the plasma membrane

I. CELL WALL (present only in Plant Cells)

- outer layer that maintains cell's shape and protects cell from mechanical damage; made of cellulose,

K. MITOCHONDRIA- powerhouse of the cell

-they contain the chemicals necessary for aerobic respiration and ATP production.

03. Prokaryotic and Eukaryotic Cells

I. INTRODUCTION

The body of all living organisms (bacteria, blue green algae, plants and animals) except viruses has cellular

II. Types of cells

A. PROKARYOTIC CELLS

The prokaryotic (Gr., pro = primitive or before; karyon = nucleus) are small, simple and most primitive.

B. EUKARYOTIC CELLS

The eukaryotic (Gr., eu =well; karyon = nucleus) cells have evolved from the prokaryotic cells and the

III. ENDOSYMBIOTIC THEORY

The endosymbiotic theory states that some of the organelles in today's eukaryotic cells were once prokaryotic

IV. Unlocking the terms

Homologous- having the same relative position, value, or structure: such as. a human arm (humerus) is

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Key terms

Histones - family of basic proteins that associate with DNA in the nucleus and help condense it into

Chromosomes - thread-like structures located inside the nucleus of animal and plant cells

04. Cell Types

I. Introduction

Cells form a working animal body through their emergent properties, which arise from successive levels of

II. TYPES OF CELL TISSUES

Key terms

Organelles - small structures that exist within cells

Organs - collections of tissues grouped together performing a common function. Organs are present not only in

05. Cell Modifications

I. Introduction

Just as viewing the body's organization from the "bottom up" (from cells to organ systems) reveals

II. What is Cell Modification?

Cell modification is the change in the structure of a cell to carry out specific functions. It enables the cell to

III. Types of Cell Modification

A. Apical Cell Modification- (Surface or luminal)

It is specialized to carry out functions that occur at these interfaces, including secretion, absorption, and

3. Microvilli- are microscopic cellular membrane protrusions that increase the

surface area for diffusion and minimize any increase in volume, and are involved in a wide variety

4. Pseudopods - is a temporary arm-like projection of a eukaryotic cell membrane that is

developed in the direction of movement. Filled with cytoplasm, pseudopodia primarily consist of

IV. Coordination and Control

- Control and coordination within a body depend on the endocrine system and the nervous system.

Key terms

Hemidesmosomes - located on the inner surface of basal plasma membranes in contact with the basal

06. Cell Cycle (Mitosis)

I. Introduction

The ability of organisms to produce more of their own kind is the one characteristic that best distinguishes living

II. Cell Division

Cell division pertains to the ability of organisms to reproduce best distinguishes living things from nonliving

A. Most cell division results in genetically identical daughter cells

The reproduction of a cell, with all of its complexity, cannot occur by a mere pinching in half; a cell is not

B. Cellular Organization of the Genetic Material

A cell's DNA, its genetic information, is called its genome.

C. Distribution of Chromosomes during Eukaryotic Cell Division

In preparation for cell division, DNA is replicated and the chromosomes condense

III. Eukaryotic Cell Division

Gametes are produced by a variation of cell division called meiosis

IV. Phases of Cell Cycle

The cell cycle consists of Mitotic (M) phase (mitosis and cytokinesis) and Interphase (cell growth and copying of

1. Prophase

Chromosomes appear as two sister chromatids held together at the centromere

2. Prometaphase

Chromosomes attach to microtubules at the kinetochores

3. Metaphase- M for middle

- All chromosomes are aligned at equator of the cell, called the metaphase plate

4. Anaphase- A for away

Proteins holding centromeres of sister chromatids are degraded, freeing individual chromosomes

5. Telophase

Chromosomes are clustered at opposite poles and decondense

V. Abnormal Cell Cycle: Cancer

Cancer is the uncontrolled growth and division of cells. Cancer cells grow and divide as long as they receive

VI. Apoptosis

Some cells in an organism are no longer needed. Apoptosis (a pup TOH sus) is a natural process of programmed

Key terms

Cell division - integral part of the cell cycle, the life of a cell from formation to its own division

Gametes - produced by a variation of cell division called meiosis

Cytoskeleton - disassembled: spindle begins to form

Chromosomes - clustered at opposite poles and decondense

Cytokinesis - well underway by late telophase

Cancer - uncontrolled growth and division of cells. Cancer cells grow and divide as long as they receive

07. Cell Cycle (Meiosis)

I. Cell Division by Meiosis: The Basis of Sex

The genetic information we inherit comes from two cells: a sperm and an egg. These cells are produced by a form

II. Meiosis Stages

A. Meiosis I reduces the chromosome number

Before cells enter meiosis, the chromosomes replicate during interphase. In prophase I, the chromosomes

B. Meiosis II begins with haploid cells

In prophase II, the unpaired chromosomes consist of two sister chromatids joined by a centromere.

III. Comparing Mitosis and Meiosis

A comparison of the events in mitosis (left) and meiosis (right). In mitosis, a diploid parent cell undergoes

V. Unlocking Vocabulary

08. Transport Mechanisms (Simple Diffusion and Facilitated Diffusion)

I. Introduction

The plasma membrane that surrounds the cell can be considered the edge of life, the boundary that separates a

II. Cellular Membranes are fluid mosaics of lipids and proteins

- Phospholipids are the most abundant lipid in the plasma membrane

A. Phospholipid bilayer

- Every cell membrane is composed of phospholipids in a bilayer. The other components of the

B. Transmembrane proteins

- A major component of every membrane is a collection of proteins that float in the lipid bilayer.

C. Interior protein network

- Membranes are structurally supported by intracellular proteins that reinforce the membrane's shape.

D. Cell-surface markers

- membrane sections assemble in the endoplasmic reticulum, transfer to the Golgi apparatus, and then

IV. Cellular membranes have an organized substructure

- It was believed that because of its fluidity, the plasma membrane was uniform, with lipids and proteins

V. Evolution of Plasma Membrane

Variations in the cell membrane lipid compositions of many species appear to be evolutionary adaptations that

A. Freshwater and Saltwater fish

Fish cells, like all cells, have semi permeable membranes. Eventually the concentration of stuff on either

B. Bacteria and Archaea

- thrive at temperatures greater than 90°C (194°F) in thermal hot springs and geysers. Their membranes

VI. Summary

Component	Composition	Function	How it Works	Example
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09. Transport Mechanisms (Simple Diffusion and Facilitated Diffusion)

I. Proteins: Multi-functional Components

Cell membranes contain a complex assembly of proteins enmeshed in the fluid soup of phospholipid molecules.

II. Proteins and protein complexes perform key functions

III. Structural features of membrane proteins relate to function

Membrane proteins can serve a variety of functions. These diverse functions arise from the diverse structures

A. The anchoring of proteins in the bilayer

Some membrane proteins are attached to the surface of the membrane by special molecules that associate

B. Transmembrane Domains

These domains are composed of hydrophobic amino acids usually arranged into helices. Proteins need only a

C. Pores

This forms a polar environment in the interior of the sheets spanning the membrane. This so-called barrel,

IV. Passive Transport Across the Membrane

Many substances can move in and out of the cell without the cell's having to expend energy. This type of

A. Transport can occur by simple diffusion

Molecules and ions dissolved in water are in constant random motion. This random motion causes a net

B. Proteins allow membrane diffusion to be selective

Many important molecules required by cells cannot easily cross the plasma membrane. These molecules

C. Facilitated diffusion of ions through channels

Because of their charge, ions interact well with polar molecules such as water, but are repelled by

D. Facilitated diffusion by carrier proteins

- Carrier proteins can help transport both ions and other solutes, such as some sugars and amino acids,

10. Transport Mechanisms (Active Transport and Bulk Vesicular Transport)

I. Osmosis

Water molecules interact with dissolved solutes by forming hydration shells around the charged solute molecules.

II. How solutes create osmotic pressure?

In a hypertonic solution, water moves out of the cell, causing the cell to shrivel. In an isotonic solution, water

III. Maintaining osmotic balance

Organisms have developed many strategies for solving the dilemma posed by being hypertonic to their

A. Extrusion

Some single-celled eukaryotes, such as the protist *Paramecium*, use organelles called contractile vacuoles to

C. Turgor

Most plant cells are hypertonic to their immediate environment, containing a high concentration of solutes in

IV. Active Transport Across Membranes

Diffusion, facilitated diffusion, and osmosis are passive transport processes that move materials down their

A. Active transport uses energy to move materials against a concentration gradient

Like facilitated diffusion, active transport involves highly selective protein carriers within the membrane that bind

B. The sodium-potassium pump runs directly on ATP

More than one-third of all of the energy expended by an animal cell that is not actively dividing is used in the

C. Coupled transport uses ATP indirectly

Some molecules are moved against their concentration gradient by using the energy stored in a gradient of a

Key terms

Active transport - one of the most important functions of any cell. It enables a cell to take up additional

Glucose - such an important molecule that there are a variety of transporters for it, one of which was discussed

11. Transport Mechanisms (Active Transport and Bulk Vesicular Transport)

I. Bulk transport across the plasma membrane

A. Exocytosis

- The cell secretes certain molecules by the fusion of vesicles with the plasma membrane.

B. Endocytosis

- The cell takes in molecules and particulate matter by forming new vesicles from the plasma membrane.

II. Types of Endocytosis

1. Phagocytosis

- A cell engulfs a particle by extending pseudopodia (singular, pseudopodium) around it and packaging it

2. Pinocytosis

- A cell continually "gulps" droplets of extracellular fluid into tiny vesicles, formed by infoldings

3. Receptor Mediated Endocytosis

- A specialized type of pinocytosis that enables the cell to acquire bulk quantities of specific

12. Structures and Functions of Biological Molecules (CHO, CHON, Enzymes and Nucleic Acids)

I. BIOLOGICAL MACROMOLECULES

Biological macromolecules are traditionally grouped into carbohydrates, nucleic acids, proteins, and lipids. In

II. Carbohydrates

They are loosely defined group of molecules that all

III. Nucleic Acids

Nucleic acids carry information inside cells, just as disks contain the information in a computer or road maps

A. Deoxyribonucleic Acid

Organisms use sequences of nucleotides in DNA to encode the information specifying the amino acid

B. Ribonucleic Acid

It carries information and is also responsible for protein synthesis. Also it is involved in control of gene

IV. Proteins

There are 20 amino acids and each amino acid is categorized into 5 chemical classes according to their R-

A. Enzyme catalysis

Enzymes are biological catalysts that facilitate specific chemical reactions. Because of this property, the

B. Defense

Other globular proteins use their shapes to "recognize" foreign microbes and cancer cells. These cell-surface

C. Transport

A variety of globular proteins transport small molecules and ions. The transport protein hemoglobin, for example,

D. Support

Protein fibers play structural roles. These fibers include keratin in hair, fibrin in blood clots, and collagen. The last

E. Motion

Muscles contract through the sliding motion of two kinds of protein filaments: actin and myosin. Contractile

F. Regulation

Small proteins called hormones serve as intracellular messengers in animals. Proteins also play many regulatory

G. Storage

Calcium and iron are stored in the body by binding as ions to storage proteins.

1. Nonpolar amino acids, such as leucine, often have R groups that contain -CH₂ or -CH₃

Key terms

Biological macromolecules - traditionally grouped into carbohydrates, nucleic acids, proteins, and lipids. In

Sugars - among the most important energy storage molecules, and they exist in several different forms

Polysaccharides - long polymers made up of monosaccharides that have been joined through

Enzymes - biological catalysts that facilitate specific chemical reactions. Because of this property, the

Calcium and iron - stored in the body by binding as ions to storage proteins

Lipids - insoluble in water because they have a high proportion of nonpolar C-H bonds

Fats - excellent energy-storage molecules. The energy stored in the C-H bonds of fats is more than

13. ATP and ADP Cycle

I. Introduction

Cells are constantly converting matter and energy from one form to another in order to get what they need to stay

II. Metabolism

Metabolism is all of a cell's chemical reactions, such as breaking down food molecules or building up muscle

III. Laws of Thermodynamics

I'm sure you use the words "work" and "energy" all the time in your daily life. In everyday terms, work is

A. The first law of thermodynamics states that energy can't be created nor destroyed

The first law of thermodynamics basically means that you can't make new energy, and you can't just make energy

III. Spontaneous and Non Spontaneous Reactions

A. Spontaneous reactions

In cellular biology, the most useful expressions of the second law of thermodynamics are those that

B. Non Spontaneous Reactions

The second law of thermodynamics says that the universe is constantly increasing in disorder. At first

Key terms

Cells - constantly converting matter and energy from one form to another in order to get what they need to stay

Metabolism - all of a cell's chemical reactions, such as breaking down food molecules or building up muscle

Work - moving something over a distance

Potential energy - energy that an object or molecule has because of its position or arrangement

Kinetic energy - energy that objects and molecules have because of their motion

Only after the spring - compressed and ready to do work does it have potential energy. Even though it's possible

14. ATP and ADP Cycle

I. Going to Work in Cellular Factory

Cells transfer energy from food molecules in order to do cellular work. You know about the work you do in your

II. ATP: The Energy Currency of Cells

The chief "currency" all cells use for their energy transactions is the nucleotide adenosine triphosphate (ATP). ATP

1. Energy from food is transferred to ATP

2. ATP provides energy for cellular work

Adenosine triphosphate (ATP) is a nucleotide, the same type of building block that is used to build DNA and RNA.

III. Enzymes: Biological Catalysts

Life is very fast paced - too fast to just wait around for necessary chemical reactions to occur, even spontaneous

Key terms

Cells - constantly building and breaking down ATP, creating an ATP/ADP cycle. Cells store energy by building ATP

Life - very fast paced - too fast to just wait around for necessary chemical reactions to occur, even spontaneous

Sodium - oxidized and becomes the sodium ion (Na⁺). Chlorine is reduced and becomes the chloride ion (Cl⁻)

Metabolism - all about getting what you need to stay alive. You know what you need on a big scale - good old

15. Photosynthesis (Part 1)

I. Introduction

Life is powered by sunshine. The energy used by most living cells comes ultimately from the Sun and is captured

II. Photosynthesis combines CO₂ and H₂O, producing glucose and O₂

Photosynthesis occurs in a wide variety of organisms, and it comes in different forms. These include a form of

II. Stages of Photosynthesis

1. capturing energy from sunlight

III. Photosynthesis and Chloroplasts

The internal membrane of chloroplasts, called the thylakoid membrane, is a continuous phospholipid bilayer

IV. Light Dependent Reactions

Each pigment molecule within the photosystem is capable of capturing photons, which are packets of energy.

IV. The Discovery of Photosynthetic Process

The story of how we learned about photosynthesis begins over 300 years ago, and it continues to this day. It starts

Key terms

Life - powered by sunshine. The energy used by most living cells comes ultimately from the Sun and is captured

Oxygenic photosynthesis - found in cyanobacteria, seven groups of algae, and essentially all land plants. These

Connections between grana - termed stroma lamellae. Surrounding the thylakoid membrane system is a

16. Photosynthesis (Part 2)

I. Introduction

You may have noticed that plants don't get around much. They're pretty much stuck where they put down their

II. Plant Structures involved in Photosynthesis

Plant structures are specialized for taking in the materials that they need:

III. Role of the Soil

Soil is useful to plants because it gives them something sturdy to stand in, it may hold water for them, and it may

IV. The Sun

Everyone knows that the Sun sends light to the Earth, but did you know that the Sun's radiation contains more

V. The Light Reaction

It shows the transfer of matter (carbon dioxide and water)

VI. Transferring Light Energy to Chemical Energy

Photosynthesis is possible because the photosynthetic pigments that absorb the light are hooked up to a little

VII. Phosphorylation

Photophosphorylation means making ATP using the energy from light.

2. The electrons from reaction center chlorophylls are excited and move to outer orbitals

VII. The Calvin cycle

The light reactions of photosynthesis transform light energy to chemical energy, producing the energy,

Key terms

Plant structures - specialized for taking in the materials that they need:

Plant leaves - full of the green pigment chlorophyll, which is very good at absorbing blue and red light, but not

Soil - useful to plants because it gives them something sturdy to stand in, it may hold water for them, and it may

Photosynthesis - possible because the photosynthetic pigments that absorb the light are hooked up to a little

Photophosphorylation - really just another way of saying "the light reactions of photosynthesis."

17. Cellular Respiration (Part 1)

I. Introduction

When you eat food, your digestive system breaks down the food into smaller and smaller pieces until it becomes a

II. What is Cellular Respiration?

-It is the breakdown of food molecules such as carbohydrates, proteins, lipids and fats.

III. Types of Cellular Respiration

A. Aerobic Respiration (with oxygen)

-stages: glycolysis (anaerobic), Krebs cycle and oxidative phosphorylation.

B. Anaerobic Respiration (without oxygen)

-products: ATP and lactic acid in animals; or ATP ethanol and CO₂ (yeast)

IV. When does the energy transfer?

During cellular respiration, electrons are removed from food molecules, transferred to the electron carrier

V. Stages of Cellular Respiration

Glycolysis is the first part of cellular respiration. During

Key terms

Glycolysis - first part of cellular respiration. During

18. Cellular Respiration (Part 2)

I. Glycolysis

The word glycolysis means "sugar splitting," and that is exactly what happens during this pathway. Glucose, a six-

II. Citric Acid Cycle

Glycolysis releases less than a quarter of the chemical energy in glucose that can be harvested by cells; most of

III. Chemiosmosis

ATP is the most useful energy carrier for cells: Whenever a cell needs to do a job, it's usually ATP that provides the

IV. Fermentation

There are two general mechanisms by which certain cells can oxidize organic fuel and generate ATP without the

Key terms

ATP - most useful energy carrier for cells: Whenever a cell needs to do a job, it's usually ATP that provides the

Electron transport chains - made of large protein complexes embedded in a membrane. They accept electrons

ATP synthase - embedded in the inner mitochondrial membrane right along with the electron transport chain. On

Fermentation - way of harvesting chemical energy without using either oxygen or any electron transport

Alcohol fermentation- pyruvate - converted to ethanol (ethyl alcohol) in two steps. The first step releases carbon

Lactic acid fermentation- pyruvate - reduced directly by NADH to form lactate as an end product, regenerating